Booklet No. 02515

2015

PAPER – I

Full marks – 100

Time – 1 hr. 15 minutes

Signature and Name of Invigilator

1. (Signature) Pramod Madhi

(Name) DR. PRAMOD MADHI

Time : 1¼ hours] **Test Booklet Code** [X] [Maximum Marks : 100

Number of Pages in this Booklet : 16 Number of Questions in this Booklet : 60

Instructions for the Candidates

1. This paper consists of sixty (60) objective type of questions, out of which the candidate would be required to answer any fifty (50) questions. In the event of the candidate attempting more than fifty questions, the first fifty questions attempted by the candidate would be evaluated.
2. At the commencement of examination, the question booklet will be given to you. In the first 5 minutes, you are required to open the booklet and compulsorily examine it as below :
   (i) To have access to the Question Booklet, tear off the paper seal on the edge of this cover page. Do not accept a booklet without sticker-seal and do not accept an open booklet.
   (ii) **Tally the number of pages and number of questions in the booklet with the information printed on the cover page. Faulty booklets due to pages/questions missing or duplicate or not in serial order or any other discrepancy should be got replaced immediately by a correct booklet from the invigilator within the period of 5 minutes. Afterwards, neither the Question Booklet will be replaced nor any extra time will be given.**
   (iii) After this verification is over, the Test Booklet Code should be entered in the OMR Sheet.
3. Each item has four alternative responses marked (A), (B), (C) and (D). You have to darken the circle as indicated below on the correct response against each item.
   Example : Ⓐ Ⓑ ● Ⓓ
   where (C) is the correct response.
4. Your responses to the items are to be indicated in the **OMR Sheet given with this Booklet, which is common for Paper I and Paper II.** If you mark at any place other than in the circle in the OMR Sheet, it will not be evaluated. **Also answer once marked shall not be cancelled or erased. If any body does it, his/her OMR sheet shall be treated as cancelled and shall not be evaluated.**
5. Read instructions given inside carefully.
6. Rough Work is to be done in the end of this booklet.
7. If you write your Name, Roll Number, Phone Number or put any mark on any part of the Answer Sheet, except for the space allotted for the relevant entries, which may disclose your identity, or use abusive language or employ any other unfair means, you will render yourself liable to disqualification.
8. You have to keep the OMR sheet with you for marking the Answer of questions in Paper II. The OMR Answer sheet is to be handed over to the invigilators at the end of the examination of Paper II compulsorily and must not carry it with you outside the Examination Hall. You are, however, allowed to carry the Test Booklets for Paper I and II with the carbon copy of OMR Sheet on conclusion of examination.
9. **Use only Ball point pen.**
10. **Use of any calculator or log table etc., is prohibited.**
11. **There is no negative marks for incorrect answers.**

SEAL

Note : There are **Sixty (60)** questions, out of which you are to answer **fifty (50)** questions only. If you answer more than **fifty (50)** questions, then only the first **fifty (50)** questions attempted will be evaluated. The questions carry **two (2)** marks each.

1. Of the following two statements one is Assertion (Ass) and the other Reason (R). Examine these two statements and decide if (Ass) and the (R) are individually true and if so whether (R) is the correct explanation for the given (Ass)

Assertion (Ass) : The Marathas emerged as the strongest native power in India after the decline of Mughal empire.

Reason (R) : The Marathas were the first to have a clear concept of a united Indian nation.

Codes :

(A) Both Ass and R are individually true and R is the correct explanation of Ass.

(B) Both Ass and R are individually true but R is not the correct explanation of Ass

(C) Ass is true but R is false

(D) Ass is false but R is true.

2. Of the following two statements one is Assertion (Ass) and the other is Reason (R). Examine these two statements and decide if Ass and R are individually true and if so whether R is the correct explanation for the given (Ass). Select your answer using Codes given below :

Assertion (Ass) : The boiling point of water decreases as the altitude increases.

Reason (R) : The atmospheric pressure increases with the increase in altitude.

Codes :

(A) Both Ass and R are individually true and R is the correct explanation of Ass.

(B) Both Ass and R are individually true but R is not the correct explanation of Ass

(C) Ass is true but R is false

(D) Ass is false but R is true.

3. Of the following two statements — one is Assertion (Ass) and the other Reason (R). Examine these two statements and decide if (Ass) and (R) are individually true and if so whether (R) is correct explanation for the given (Ass). Select the correct answer from the codes given below :

Assertion (Ass) : A piece of copper and a piece of tinted glass are heated to the same temperature. When touched thereafter the copper piece appears hotter than the glass piece.

Reason (R) : The density of the copper is more than that of glass.

Codes :

(A) Both Ass and R are individually true and R is the correct explanation of Ass.

(B) Both Ass and R are individually true but R is not the correct explanation of Ass

(C) Ass is true but R is false

(D) Ass is false but R is true.

4. Of the following, the main purpose of state certification of teachers is to

(A) Monitor the quality of teacher training institutions.

(B) Provide for a uniform standard of entry-level teacher competency throughout the state.

(C) Exclude from the profession those who are not trained in pedagogy.

(D) Exclude from the profession those who are mentally unhealthy.

5. A successful teacher would do well to avoid the following. Express your responses as 'True' or 'False' using Codes given below :

a) Stating opinions as facts.

b) Under promise his performance goals and over perform.

c) While interacting with outsiders does not reveal too much about his Head of institution

d) Avoids admitting ignorance.

Codes	a	b	c	d
(A)	False	True	True	False
(B)	True	False	True	True
(C)	True	False	False	True
(D)	False	False	True	False

6. The following statements describe certain characteristics of a teacher who is trying to succeed in the profession. Which of these are 'True' and which are 'False'. Indicate using codes.

a) Admits if something wrong is said or done.

b) Feedback from students is not sought.

c) Takes pains to connect with students.

d) Does not care about his style of work.

Codes	a	b	c	d
(A)	True	False	True	False
(B)	True	True	False	False
(C)	True	True	True	False
(D)	False	True	True	False

7. Match the terms given under Column I with the corresponding meanings given under Column II and indicate your responses using codes given below :

Column I	Column II
a) Data	1. Basis to make prediction of outcome of actions
b) Information	2. Processed information usable as formal / official record.
c) Knowledge	3. Basic facts and entities
d) Document	4. Meaningful matter usable in decision making.

Codes	a	b	c	d
(A)	3	2	1	4
(B)	3	4	1	2
(C)	4	3	2	1
(D)	3	1	4	2

8. Match the names of certain important days given under Column I with the corresponding dates, given under Column II, on which they are observed / celebrated, using codes given below :

Column I	Column II
a) Teachers' Day	1. November 14
b) International Literacy Day	2. June 5
c) Children's Day	3. September 5
d) World Environment Day	4. September 8

Codes	a	b	c	d
(A)	4	3	2	1
(B)	3	4	1	2
(C)	3	2	1	4
(D)	2	4	1	3

9. Criticism referring to the authenticity of data source is called —

(A) Empirical Criticism

(B) Constructive Criticism

(C) External Criticism

(D) None of the above

10. Choose the appropriate alternative given in the codes to replace the question mark

Pleader – Client, Examiner – Examinee, Preceptor – ?

(A) Path-finder

(B) Perceiver

(C) Customer

(D) Disciple

11. Which one of the following is an indication of the quality of a research journal?

(A) g-index

(B) h-index

(C) i/o-index

(D) Impact factor

12. When a teacher interacts and communicates with the students in a classroom the communication type is known as

(A) Face to face Communication

(B) Group Communication

(C) Mass Communication

(D) Interpersonal Communication

13. Interaction within the classroom should generate

(A) Information

(B) Ideas

(C) Argument

(D) Controversy

14. Find out the missing number of a series — 5, 12, 27, 58, ?

(A) 112

(B) 116

(C) 121

(D) 127

15. The extent to which the findings of research can be generalised is called

(A) External Validity

(B) Internal Validity

(C) Content Validity

(D) None of the above

16. Independent variables are manipulated in —

(A) Ex post facto research

(B) Experimental research

(C) Ethnographic research

(D) Historical research

17. Which of the following is a non-probability sampling?

(A) Stratified Random Sampling

(B) Cluster Sampling

(C) Systematic sampling

(D) Snowballing

18. A percentile rank indicates a pupil's

(A) Relative position in a group

(B) Percentage of marks in end term exam.

(C) Percentage of marks in a specific subject

(D) None of the above

19. Which of the following is the correct sequence for writing a thesis ?

(A) statement of the problem → sample → objectives → analysis → significance of the study → interpretation

(B) objectives → sample → statement of the problem → significance of the study → analysis → interpretation

(C) statement of the problem → objectives → sample → analysis → interpretation → significance of the study

(D) objectives → statement of the problem → significance of the study → sample → analysis → interpretation

20. Match Set A with Set B

Set A :

(i) Teacher researchers recti difficulties in teachin mathematics in Class VI

(ii) A researcher administers pre test and post test an gives a treatment in betwee

(iii) A researcher administer an achievement test on sample & generalizes abou the population

(iv) A researcher looks u manuscripts to find ou about a great educationis

Set B :

(a) experimental research
(b) survey research
(c) historical research
(d) action research

Use Codes below to point th correct match

Codes	i	ii	iii	iv
(A)	a	b	c	d
(B)	d	c	b	a
(C)	d	a	b	c
(D)	b	d	a	c

21. Mass Communication

(A) is meant for deliverin messages to the resident of a city.

(B) does not need technology based media vehicles.

(C) is an impersonal tool fo delivering information t the masses.

(D) is more effective tha dyadic communication s far as rich is concerned.

22. Find the number that comes next in the sequence

8, 15, 24, 35 ... 48

(A) 37

(B) 42

(C) 48

(D) 35

23. Informal Communication network within the organisation in known as

(A) interpersonal communication

(B) intrapersonal communication

(C) mass communication

(D) grapevine communication

24. Find the missing word in the following sentence from the choices given below :

_____ is the meaningful outcome of processed data.

(A) Knowledge

(B) Information

(C) Output

(D) Summary

25. Statistical observation arranged in chronological order is called—

(A) Regression

(B) Progression

(C) Time Series

(D) Logitudinal research

Read the following passage and answer the **five** questions that follow :

Many traditional Japanese beliefs and practices hark back to prehistoric customs and most of these form the core of Shinto, the only major religion indigenous to Japan. Indian Buddhism, the Chinese contributions of Confucianism and Taoism (transmitted first through the cultural bridge of Korea), and, much later, Christianity were introduced to Japan from outside. All these foreign traditions have undergone significant transformations in a process of mutual influence with the native tradition.

In Judaism and Christianity religion entails faith in one supreme deity; revelation of the will of this deity in a sacred book; concern with sin as disobedience to the deity; relation of man to divinity through a conscious decision or act of faith; specific ecclesiastical organisations, involving regular attendance and worship; and ethical behaviour linked directly to this religious commitment.

Japanese religion differs significantly on each of the aforementioned points; there are not one but many deities, there is no one sacred book, but many religious scriptures; rather than emphasis on sin as disobedience to the deity there is concern with ritual impurity and purification. One person usually participates in more than one religious tradition; there is no regular

worship day comparable to the Sabbath but many seasonal festivals; and ethical codes are more closely related to family life and philosophy than to organised religion, while ethical shortcomings are not related directly to divine will but are considered in terms of human imperfection.

26. How is sin interpreted in Judaism and Christianity?

(A) Thought and action taken against other beings.

(B) Violation of socio-cultural norms.

(C) Disobedience to supreme deity.

(D) Deviant behaviour causing hurt to others.

27. What is the status of religious books / scriptures in Japanese religion?

(A) A single sacred book followed

(B) Several religious scriptures followed

(C) No sacred book

(D) None of the above.

28. What is the important day of worship according to Japanese religion?

(A) Wednesday

(B) No day prescribed

(C) Any day of the week

(D) All seasonal festival days.

29. Which is the major religious tradition of Japan?

(A) Shinto

(B) Confucianism

(C) Buddhism

(D) Taoism

30. How are ethical shortcomings explained in Japanese religion?

(A) As isolation against religious authority.

(B) Thoughts against divine will

(C) As human imperfections

(D) All of the above

31. Identify the pattern in the following sequence and choose the number that will come next.

6, 19, 58, 175, 526, ?

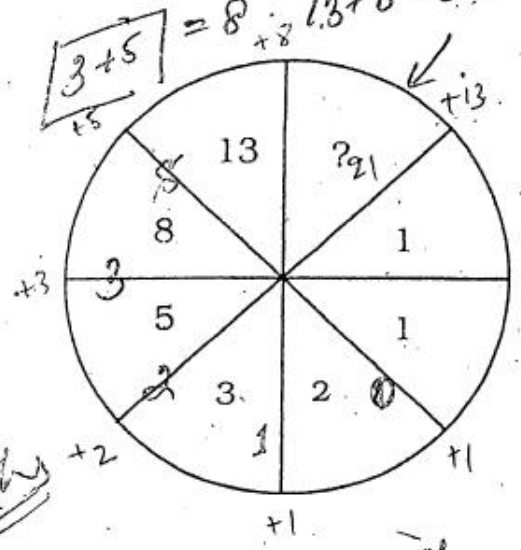
(A) 1669

(B) 1579

(C) 1578

(D) 1668

32. Choose the correct number to fill up the blank



(A) 13

(B) 22

(C) 2

(D) 21

33. The maximum number of triangle with

3 points as vertices is 1

4 points as vertices is 4

5 points as vertices is 10

What is the maximum number of triangle with 6 points as vertices :



(A) 20

(B) 25

(C) 30

(D) 40

34. The missing number in the following sequence is

1, 9, 25, 49, ☐, 121

(A) 64

(B) 81

(C) 100

(D) 121

35. In a class of 50 students, all study Mathematics or English, or both. The Mathematics class consists of 25 students, and the English class consists of 30 students. Is this possible ?

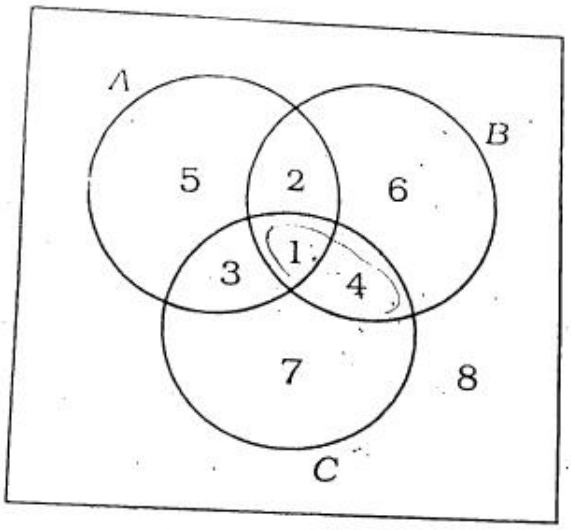
(A) Not possible, because 25+30=55, which is more than 50.

(B) Possible, because there may be students who come from other classes.

(C) Not possible, because there should be 25 students in the English class.

(D) Possible, because some students have been added twice.

36. Consider the Venn diagram of Sets A, B and C in *figure*-1. Observe that these sets partition the rectangle (universal set) into 8 numbered sets denoted by 1, 2, 3, 4, 5, 6, 7 and 8. Match the following relation of two columns and find the correct answer from the codes given below :



figure–1

Column-1	Column-2
a) $A \cap B^c \cap C^c$	1. 6
b) $A^c \cap B \cap C^c$	2. 7
c) $A^c \cap B^c \cap C$	3. 8
d) $A^c \cap B^c \cap C^c$	4. 5

Codes	a	b	c	d
(A)	2	3	4	1
(B)	3	4	1	2
(C)	4	1	2	3
(D)	1	2	3	4

37. In an examination 25% fails in Mathematics, 15% fails in Physics and 5% fails in both. The number of students that passes in both the subjects is 130. The total member of students appeared in the examination is

(A) 200
(B) 300
(C) 350
(D) 400

38. The sum of the series $1+2+3+\ldots\ldots+5$ is

(A) 35
(B) 45
(C) 55
(D) 65

39. The marks secured by 10 students in a class are : 38, 70, 48, 40, 42, 55, 63, 46, 54, 44 and the median of the distribution is 47. The number of students in percentage securing above the median is

(A) 50%
(B) 40%
(C) 30%
(D) 20%

40. Four students W X Y Z appeared in two papers I and II in the Test. Their scores out of 100 are given below :

	I	II
W	80	62
X	73	71
Y	90	50
Z	56	85

Who has obtained the highest average ?

(A) W

(B) X

(C) Y

(D) Z

41. The number of students in a school is given below :

	Boys	Girls
Class VI	56	62
Class X	58	29

Select True or False statements for the following :

(i) The numbers of boys in Class VI & X are nearly equal.

(ii) The number of girls has decreased considerably in Class X.

(iii) The number of boys and girls is nearly the same in Class X.

(iv) Girls tend to stay on and complete schooling.

Codes	i	ii	iii	iv
(A)	False	True	False	True
(B)	True	True	False	False
(C)	True	True	False	True
(D)	True	False	False	False

42. Twelve people form a club. By picking lots, any one of them will host a dinner for all once in a month. The number of dinners a given member has to host in one year is

(A) one

(B) zero

(C) two

(D) cannot be predicted.

43. Full form of WWW is

(A) World Wide Network

(B) Wide World Network

(C) World Wide Web

(D) Wide Web Network

44. A computer peripheral device that is used to read printed / handwritten / type written characters into machine encoded text thus enabling us to get the digitized text is

(A) OMR

(B) OCR

(C) Digital Camera

(D) Disk Driver

45. Which of the following is not a web browser ?

(A) Internet Explorer

(B) Mozilla Firefox

(C) Yahoo

(D) Chrome

46. Which of the following is the least pollutant ?

(A) Carbon monoxide

(B) Sulphur-di-oxide

(C) Nitrogen-di-oxide

(D) Carbon-di-oxide

47. Of the statements given below some are true and some are false. Find the correct statements from the codes below :

a) One common gaseous air pollutant is Chlorofluoro-carbons (CFCs).

b) UNEP — United Nations Education Programme

c) The Chairman of the National Knowledge Commission was Sam Pitroda.

d) It was Charles Darwin who coined the phrase "Survival of the fittest".

Codes	a	b	c	d
(A)	True	False	True	False
(B)	False	True	False	True
(C)	True	True	False	False
(D)	False	False	False	True

48. Match the following physical features to the geological structures they are related to and select the correct match from the table given below

1. Horst and graben	a. Thrust faults
2. Schuppen structure	b. Anticlines
3. Saddle reef	c. Subduction zones
4. Trenches	d. Normal faults

Codes	1	2	3	4
(A)	a	b	c	d
(B)	d	a	b	c
(C)	b	d	a	c
(D)	c	a	d	b

49. Match the following types of sedimentary deposits to their environment of deposition and select the correct match from the table given below

1. Aeolian deposits	a. Lake
2. Limnic deposits	b. Desert
3. Paludeal deposits	c. Flood plain
4. Fluvial deposits	d. Swamp

Codes	1	2	3	4
(A)	a	b	d	c
(B)	c	d	a	b
(C)	a	c	b	d
(D)	b	a	d	c

50. Match the following river dams to the States they belong and select the correct match from the table given below :

1.	Barapani Dam	a.	Maharastra
2.	Koina Dam	b.	West Bengal
3.	Damodar Valley Project	c.	Orissa
4.	Hirakud Dam	d.	Karnataka
		e.	Meghalaya

Codes	1	2	3	4
(A)	a	c	d	c
(B)	e	a	b	c
(C)	c	d	a	b
(D)	b	a	c	d

51. Following are two statements — one Assertion (Ass) and the other Reason (R). Find the correct answer from the codes below '

Assertion (Ass) : Sea floor spreading hypothesis asserts that older and older portion of the sea floor move continuously further and further away from the mid-oceanic ridges.

Reason (R) : There is continuous injection of magma along the ridges.

Codes :-

(A) (Ass) is true and (R) is the true explanation of it.

(B) (Ass) is true but (R) is not the explanation of (Ass)

(C) (Ass) is not true but (R) is true.

(D) Both (Ass) and (R) are not true.

52. Following are two statements — one Assertion (Ass) and the other Reason (R). Find the correct answer from the codes below

Assertion (Ass) : Unlike the hilly terrain around it, the Brahmaputra Valley, in spite of frequent flood havocs, is thickly populated by human beings.

Reason (R) : The alluvial soil deposited by the river is fertile and supports good cultivation.

Codes :

(A) (Ass) is true but (R) is not the correct explanation of it.

(B) (Ass) is true and (R) is the correct explanation of it.

(C) (Ass) is not true but (R) is true.

(D) Both (Ass) and (R) are not true.

53. Following are two statements — one Assertion (Ass) and the other Reason (R). Find the correct answer from the codes below

Assertion (Ass) : Volcanic eruptions issues highly toxic gases which cause death to many animals but standing human beings are usually spared of death.

Reason (R) : These gases are heavier than air and are spread at low levels along the surface of the earth.

Codes :

(A) (Ass) is correct but (R) is not the true explanation.

(B) (Ass) is not true, while (R) is true.

(C) (Ass) is true and (R) is the correct explanation to it.

(D) Both (Ass) and (R) are not true.

54. Which of the following socio-economic problems are related with over population

(A) food

(B) employment

(C) good fertility of soil

(D) all of these

55. For the first time environment was put as an Agendum of International Concern in the Stockholm Conference in the year of

(A) 1972

(B) 1975

(C) 1982

(D) 1985

56. Chipko movement is

(A) a movement of political struggle

(B) an environmental movement

(C) a movement for independence

(D) a movement against corrupt political leaders.

57. Literary speaking the term "environment" means

(A) surroundings

(B) human beings

(C) natural phenomenon

(D) landscape

58. Lithosphere means

(A) Layer of gases

(B) Layer of sands

(C) Layer of gravels

(D) None of these

59. The main source of energy in an ecosystem is

(A) heat liberated during respiration

(B) heat liberated during fermentation

(C) sugar stored in plants

(D) sunlight (solar energy)

60. Which of the following are physiological manifestations of sound pollution.

(A) decrease of heart output

(B) damage of night vision

(C) fluctuation in blood pressure

(D) all of these